Practical Demo: Constructing a Benchmark-Aware Bond Portfolio

From cash to a cell-matched portfolio and a multi-factor portfolio

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The investment decision

A manager receives **$50 million in cash and owns no bonds**. Construct a benchmark-aware bond portfolio in two ways:

1. **Cell-based construction:** match the benchmark's sector/maturity buckets, then select bonds within each bucket at the lowest acquisition cost.
2. **Multi-factor construction:** choose bond weights that minimize predicted tracking error using Treasury key-rate and spread-factor exposures.

Both methods use the same benchmark, investable universe, cash budget, and portfolio limits. The cell method additionally requires exact cell weights. There are no starting bond holdings, sales, or rebalancing turnover penalties. Every nonzero position is an initial purchase.

All bond descriptions, analytics, spreads, costs, and risk estimates below are **illustrative inputs**, not market observations or executable quotes. Weights refer to the bond sleeve after purchase costs; all amounts are in US dollars.

1 Specify the mandate

Requirement	Limit
Initial cash, including acquisition costs	$50 million
Portfolio positions	Long-only; bond-sleeve weights sum to 100%
Maximum weight in one bond	18% of the bond sleeve
Minimum Treasury weight	25%
Maximum corporate weight	50%
Duration versus benchmark	Within 0.05 years
Weighted yield proxy versus benchmark	At least 2 bp higher
Acquisition cost	At most 6 bp of invested bond market value

The modest yield-proxy floor gives both constructors a deliberate active objective. It is not a forecast of excess return. No hard tracking-error limit is imposed: both portfolios are evaluated with the same model, and the multi-factor constructor minimizes that measure.

1.1 Account for purchase costs inside the cash budget

Let B be initial cash, w_i bond-sleeve weights, and c_i each bond's purchase cost per dollar of market value. The amount invested in bonds is

$$Q = \frac{B}{1 + \sum_i w_i c_i}.$$

Buy Qw_i of each bond and pay $Qw_i c_i$ of acquisition costs. Therefore

$$\sum_i Qw_i + \sum_i Qw_i c_i = B.$$

No additional cash is assumed. Costs reduce the initial amount invested; they are a one-time wealth drag, not annual tracking-error volatility.

2 Define the investable bonds and benchmark

The benchmark is an explicitly supplied set of bond weights. It is a reference index, not an existing client portfolio. Both constructors may buy any of the 12 securities below. Each corporate label represents a different issuer, so bond concentration and issuer concentration coincide within this universe.

All bonds are illustrative dollar-denominated, option-free, fixed-coupon bonds with semiannual payments and $100 face value. Settlement is immediately after a coupon date, so accrued interest is zero. Coupons are supplied inputs; prices, yields, durations, and convexities are calculated from those cash flows.

The Treasury zero curve has rates of **4.15%, 4.35%, and 4.55% at 2, 5, and 10 years**, quoted annually with semiannual compounding. Zero rates interpolate linearly between nodes and stay flat before 2 years. Every payment is within 10 years. The supplied spread is a constant addition to this zero curve, in basis points under the same compounding convention; Treasury spreads are zero.

Individual yields to maturity are quoted annually with semiannual compounding. The weighted yield remains a proxy, not the portfolio cash-flow internal rate of return. The weighted average zero-curve spread is likewise a proxy, not a separately solved portfolio spread.

2.1 Benchmark weights and classification

Bond	Sector	Maturity (years)	Benchmark (%)
UST 2Y	Treasury	2	10.00
UST 5Y	Treasury	5	13.00
UST 10Y	Treasury	10	10.00
Agency 3Y	Agency	3	9.00
Agency 7Y	Agency	7	10.00
Bank A 3Y	Corporate	3	7.00
Bank B 5Y	Corporate	5	9.00
Utility A 7Y	Corporate	7	8.00
Industrial A 10Y	Corporate	10	7.00
Industrial B 5Y	Corporate	5	7.00
Telecom A 10Y	Corporate	10	5.00
Consumer A 7Y	Corporate	7	5.00

2.2 Yields, spreads, and acquisition costs

Purchase costs are paid once on the acquired market value, beyond that market value. They include the assumed execution friction; no additional spread or fee is added elsewhere in the budget.

Bond	Yield (%)	Spread (bp)	Trading cost (bp)
UST 2Y	4.15	0	1
UST 5Y	4.34	0	2
UST 10Y	4.52	0	3
Agency 3Y	4.59	38	3
Agency 7Y	4.96	55	5
Bank A 3Y	5.13	92	6
Bank B 5Y	5.46	112	7
Utility A 7Y	5.79	138	9
Industrial A 10Y	6.16	165	12
Industrial B 5Y	5.59	125	8
Telecom A 10Y	6.33	182	14
Consumer A 7Y	5.89	148	10

Bond	Coupon (%)	Full price
UST 2Y	4.150	100.000
UST 5Y	4.350	100.049
UST 10Y	4.550	100.257
Agency 3Y	4.420	99.518
Agency 7Y	4.700	98.469
Bank A 3Y	5.050	99.771
Bank B 5Y	5.280	99.236
Utility A 7Y	5.550	98.645
Industrial A 10Y	5.850	97.725
Industrial B 5Y	5.400	99.197
Telecom A 10Y	6.000	97.604
Consumer A 7Y	5.620	98.481

2.3 Treasury and spread sensitivities

Key-rate durations and spread durations are measured in years. Convexity is measured in years squared. These are **calculated sensitivities**, not assigned allocations of a supplied duration.

For cash flows $CF_{i,t}$, the price per \$100 face is

$$P_i = \sum_t CF_{i,t} \left(1 + \frac{z(t) + s_i}{2}\right)^{-2t}.$$

A key-rate shock moves one zero-curve node and linearly interpolates that move between neighboring nodes, holding the other node quotes fixed. The 2Y node also controls rates before 2 years. The three unit bump functions sum to one, so the three key-rate durations add to parallel zero-curve duration.

The **2Y Treasury has zero 5Y and 10Y sensitivity**: neither bump changes any rate used to discount its cash flows. Every bond maturing at 5 years or earlier has zero 10Y sensitivity. A 3Y bond can have 5Y sensitivity because its later cash flows fall between the 2Y and 5Y nodes. A 5Y bond can have 2Y sensitivity through its earlier coupons.

For these option-free bonds, shifting the full Treasury zero curve or the bond's constant spread has the same discount-rate effect. Non-Treasury spread duration therefore equals parallel rate duration in this model. Treasury spread sensitivity is zero because Treasury spreads are held at zero.

Bond	KRD 2	KRD 5	KRD 10	Spread duration
UST 2Y	1.90	0.00	0.00	0.00
UST 5Y	0.25	4.20	0.00	0.00
UST 10Y	0.26	0.77	6.93	0.00
Agency 3Y	1.89	0.88	0.00	2.78
Agency 7Y	0.27	3.54	2.08	5.89
Bank A 3Y	1.88	0.87	0.00	2.75
Bank B 5Y	0.29	4.04	0.00	4.34
Utility A 7Y	0.31	3.44	1.97	5.72
Industrial A 10Y	0.32	0.92	6.19	7.44
Industrial B 5Y	0.30	4.03	0.00	4.32
Telecom A 10Y	0.33	0.93	6.12	7.38
Consumer A 7Y	0.31	3.43	1.96	5.70

Bond	Duration	Convexity
UST 2Y	1.90	4.61
UST 5Y	4.45	23.18
UST 10Y	7.96	75.83
Agency 3Y	2.78	9.33
Agency 7Y	5.89	40.89
Bank A 3Y	2.75	9.19
Bank B 5Y	4.34	22.31
Utility A 7Y	5.72	39.17
Industrial A 10Y	7.44	68.71
Industrial B 5Y	4.32	22.20
Telecom A 10Y	7.38	67.96
Consumer A 7Y	5.70	39.01

3 Identify the four risk factors

A **factor is a market change**. A duration is a security's **sensitivity** to that change. The risk model contains exactly four common factors:

Factor	Change being modeled	Bond sensitivity
2Y Treasury yield	Change at the 2-year Treasury node	2Y key-rate duration
5Y Treasury yield	Change at the 5-year Treasury node	5Y key-rate duration
10Y Treasury yield	Change at the 10-year Treasury node	10Y key-rate duration
Common spread	Change in the common corporate-spread driver	Spread duration times sector beta

The spread-driver beta is **0 for Treasuries, 0.25 for agencies, and 1 for corporates**. A +100 bp common-spread shock therefore means unchanged Treasury spreads, +25 bp agency spreads, and +100 bp corporate spreads. This same scaling is used in both predicted risk and scenarios.

There is no separate level factor on top of the three key rates. A parallel Treasury move means applying the same change to all three nodes; adding another level shock would count that move twice.

3.1 Return mapping and units

For changes measured in basis points, the first-order bond return in basis points is

$$r_i \approx -KRD_{i,2}\Delta y_2 - KRD_{i,5}\Delta y_5 - KRD_{i,10}\Delta y_{10} - \beta_i SD_i \Delta s + \epsilon_i.$$

For example, the 5Y Treasury has 5Y key-rate duration **4.203 years**. A 10 bp increase at that node implies a first-order loss of approximately **42.03 bp**. Bank B's spread duration is **4.337 years**, implying approximately **43.37 bp** of loss for a 10 bp common-spread increase with Treasury yields unchanged.

Here ϵ_i is security-specific return. Residuals are assumed mutually independent and independent of all four common factors. The model omits additional rating, sector, liquidity, currency, and option factors.

3.2 Factor volatilities, correlations, and residual risk

All volatilities are **annualized**, and predicted tracking error is annualized. Factor volatilities below are volatilities of yield/spread changes in bp per square-root year; residual volatilities are bond-return volatilities in the same time units. They are supplied inputs, not estimates from the bond table.

Factor	Annual shock volatility (bp)
2Y Treasury yield	70.0
5Y Treasury yield	80.0
10Y Treasury yield	90.0
Common spread	60.0

Factor	2Y	5Y	10Y	Spread
2Y	1.00	0.82	0.62	0.05
5Y	0.82	1.00	0.85	0.08
10Y	0.62	0.85	1.00	0.10
Spread	0.05	0.08	0.10	1.00

Residual group	Annual residual return volatility (bp)
Treasury / agency	30
Corporate	100

The factor covariance matrix is $\Sigma_F = \mathrm{diag}(\sigma_F)\rho\,\mathrm{diag}(\sigma_F)$, in squared basis points. Correlations are assumed fixed for this exercise. In practice they require a separate historical or forward-looking risk-estimation process; security terms and benchmark weights alone do not determine them.

4 Map holdings into active risk

Let L contain the negative duration loadings in the return equation and let $a = w - w_B$ be active bond-sleeve weights. Predicted active-return variance is

$$TE^2 = (a'L)\Sigma_F(L'a) + \sum_i a_i^2\sigma_{\epsilon,i}^2.$$

The square root is annual TE in basis points. For example, increasing one bond's weight by 10 percentage points adds one tenth of its factor sensitivities and residual exposure, before accounting for the offsetting funding position.

This comparison measures the two normalized bond sleeves against the same normalized benchmark. Acquisition costs are reported separately as a known initial drag. The benchmark index itself has no acquisition cost.

5 Construct the two portfolios

5.1 Method 1: exact cell matching

Define cells by **sector** and **remaining maturity**. Sectors are Treasury, agency, and corporate; maturity buckets are $(0, 4]$, $(4, 8]$, and over 8 years. There are nine possible cells and eight occupied cells in this universe.

Cell	Benchmark %
Agency / 0-4Y	9.00
Agency / 4-8Y	10.00
Corporate / 0-4Y	7.00
Corporate / 4-8Y	29.00
Corporate / 8Y+	12.00
Treasury / 0-4Y	10.00
Treasury / 4-8Y	13.00
Treasury / 8Y+	10.00

For each occupied cell g, require

$$\sum_{i \in g} w_i = \sum_{i \in g} w_{B,i}.$$

Within those cell budgets, minimize $\sum_i c_i w_i$ subject to all common mandate limits. Duration and yield are guardrails; **the selection step does not use factor covariance or predicted tracking error**. A linear program makes the cheapest feasible selection within the exact cells. Its output is assessed with the shared risk model afterward.

5.2 Method 2: multi-factor risk minimization

Remove the exact-cell constraints and instead minimize the common model's predicted tracking-error variance:

$$\min_{w} \; TE(w)^2.$$

Keep every common mandate limit, including duration, concentration, sector, yield-proxy floor, and acquisition-cost cap. The optimizer can shift between cells when that improves the combined factor and residual risk. The benchmark itself has zero TE but fails the required 2 bp yield-proxy increment.

Both problems use long-only weights summing to one. Dividing risk variance by 10,000 in the calculation is only numerical scaling; it does not change the solution. The initial feasible point is a cheapest feasible purchase plan, not a set of pre-existing holdings.

6 Compare the constructed portfolios

6.1 Bond allocations

All portfolio weights below are percentages of their invested bond sleeve. A zero weight means the constructor does not buy that security.

Bond	Benchmark (%)	Cell (%)	Multi-factor (%)
UST 2Y	10.00	10.00	7.68
UST 5Y	13.00	13.00	9.95
UST 10Y	10.00	10.00	9.66
Agency 3Y	9.00	9.00	12.10
Agency 7Y	10.00	10.00	12.97
Bank A 3Y	7.00	7.00	7.39
Bank B 5Y	9.00	0.00	8.90
Utility A 7Y	8.00	2.04	7.64

Bond	Benchmark (%)	Cell (%)	Multi-factor (%)
Industrial A 10Y	7.00	12.00	6.45
Industrial B 5Y	7.00	12.41	7.30
Telecom A 10Y	5.00	0.00	5.01
Consumer A 7Y	5.00	14.55	4.95

6.2 Verify cell matching

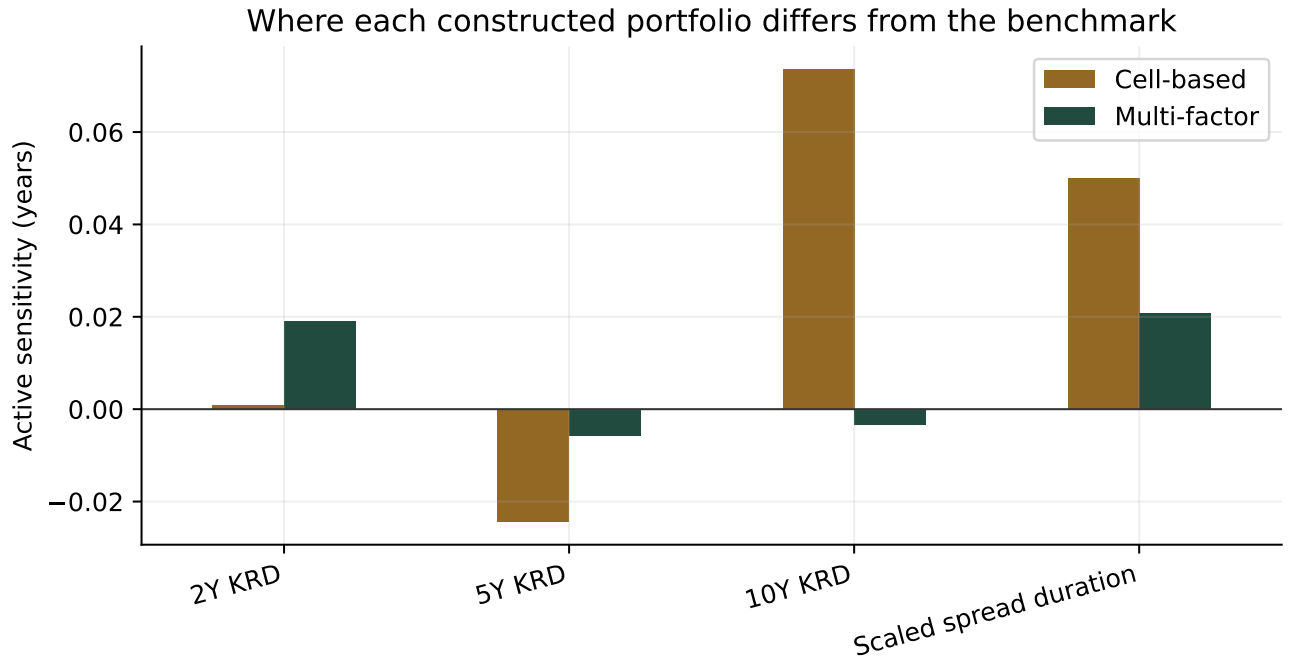
Cell	Benchmark %	Cell %	Factor %
Agency / 0-4Y	9.00	9.00	12.10
Agency / 4-8Y	10.00	10.00	12.97
Corporate / 0-4Y	7.00	7.00	7.39
Corporate / 4-8Y	29.00	29.00	28.79
Corporate / 8Y+	12.00	12.00	11.46
Treasury / 0-4Y	10.00	10.00	7.68
Treasury / 4-8Y	13.00	13.00	9.95
Treasury / 8Y+	10.00	10.00	9.66

6.3 Compare exposure and risk

Portfolio	Yield proxy (%)	Duration (years)	Annual TE (bp)
Benchmark	5.087	4.920	0.000
Cell-based	5.107	4.970	18.028
Multi-factor	5.107	4.930	2.454

Portfolio	Treasury (%)	Corporate (%)	Largest bond (%)
Benchmark	33.00	48.00	13.00
Cell-based	33.00	48.00	14.55
Multi-factor	27.30	47.63	12.97

Portfolio	Average spread (bp)
Benchmark	73.28
Cell-based	75.02
Multi-factor	75.24



Exact cell matching does not imply exact factor matching. Bonds in one cell can have different key-rate durations, spread sensitivities, and residual risk. The multi-factor result has the smallest modeled TE among portfolios satisfying the common mandate. This does not establish higher future realized returns or robustness to a misspecified covariance model.

6.4 Explain the sources of predicted risk

The following amounts add in **variance**, not standard deviations. The factor contribution for factor k is $z_k(\Sigma_F z)_k$, where $z = a'L$. A contribution can be negative when covariance offsets other active risks.

Portfolio	Source	Variance (bp²)
Cell-based	2Y Treasury yield	0.152
Cell-based	5Y Treasury yield	-7.750
Cell-based	10Y Treasury yield	35.128
Cell-based	Common spread	10.527
Cell-based	Independent residuals	286.943
Multi-factor	2Y Treasury yield	1.104
Multi-factor	5Y Treasury yield	-0.216
Multi-factor	10Y Treasury yield	-0.078
Multi-factor	Common spread	1.539
Multi-factor	Independent residuals	3.674

7 Turn initial cash into purchase orders

These are **purchases from cash**, not differences from an existing bond portfolio. Bond market values plus acquisition costs must equal $50 million.

Market value is the full settlement value, including accrued interest. Converting it to face amount requires the full price per $100 face, feasible lots, and available inventory. The calculated model prices are not executable quotes, and no lot-size or inventory assumptions are supplied, so the table remains a market-value allocation plan rather than executable face-amount tickets.

Cell-based initial purchase plan

Bond	Weight (%)	Buy $(m)\|Cost()$	
UST 2Y	10.000	4.997	499.704
UST 5Y	13.000	6.496	1299.231
UST 10Y	10.000	4.997	1499.112
Agency 3Y	9.000	4.497	1349.201
Agency 7Y	10.000	4.997	2498.521
Bank A 3Y	7.000	3.498	2098.757
Bank B 5Y	0.000	0.000	0.000
Utility A 7Y	2.042	1.020	918.190
Industrial A 10Y	12.000	5.996	7195.739
Industrial B 5Y	12.410	6.201	4961.009
Telecom A 10Y	0.000	0.000	0.000
Consumer A 7Y	14.549	7.270	7269.947

Multi-factor initial purchase plan

Bond	Weight (%)	Buy $(m)\|Cost()$	
UST 2Y	7.682	3.839	383.889
UST 5Y	9.953	4.974	994.746
UST 10Y	9.663	4.829	1448.643
Agency 3Y	12.098	6.046	1813.683
Agency 7Y	12.969	6.481	3240.272
Bank A 3Y	7.387	3.691	2214.731
Bank B 5Y	8.897	4.446	3112.108
Utility A 7Y	7.638	3.817	3435.231
Industrial A 10Y	6.449	3.223	3867.325
Industrial B 5Y	7.302	3.649	2919.018
Telecom A 10Y	5.007	2.502	3502.560
Consumer A 7Y	4.954	2.476	2475.628

Method	Bonds ()$\|Costs()$	Cash used ($)	
Cell-based	49,970,410.59	29,589.41	50,000,000.00
Multi-factor	49,970,592.17	29,407.83	50,000,000.00

8 Stress the completed portfolios

A parallel Treasury shock changes all three zero-curve nodes equally. A twist changes them by different amounts. For a parallel rate change, duration and convexity provide the familiar approximation:

$$\Delta P_i/P_i \approx -D_i\Delta y - \beta_i SD_i\Delta s + \tfrac{1}{2}C_i(\Delta y)^2.$$

The **scenario tables and interactive controls reprice every cash flow** using the shocked zero curve and spread. They do not switch approximations when the curve changes shape. Full repricing includes rate and spread convexity and their interaction under this fixed-cash-flow model.

The scenarios are instantaneous price changes, excluding carry, defaults, residual-return shocks, and reinvestment. No probabilities are assigned.

8.1 Gross price changes and wealth changes after costs

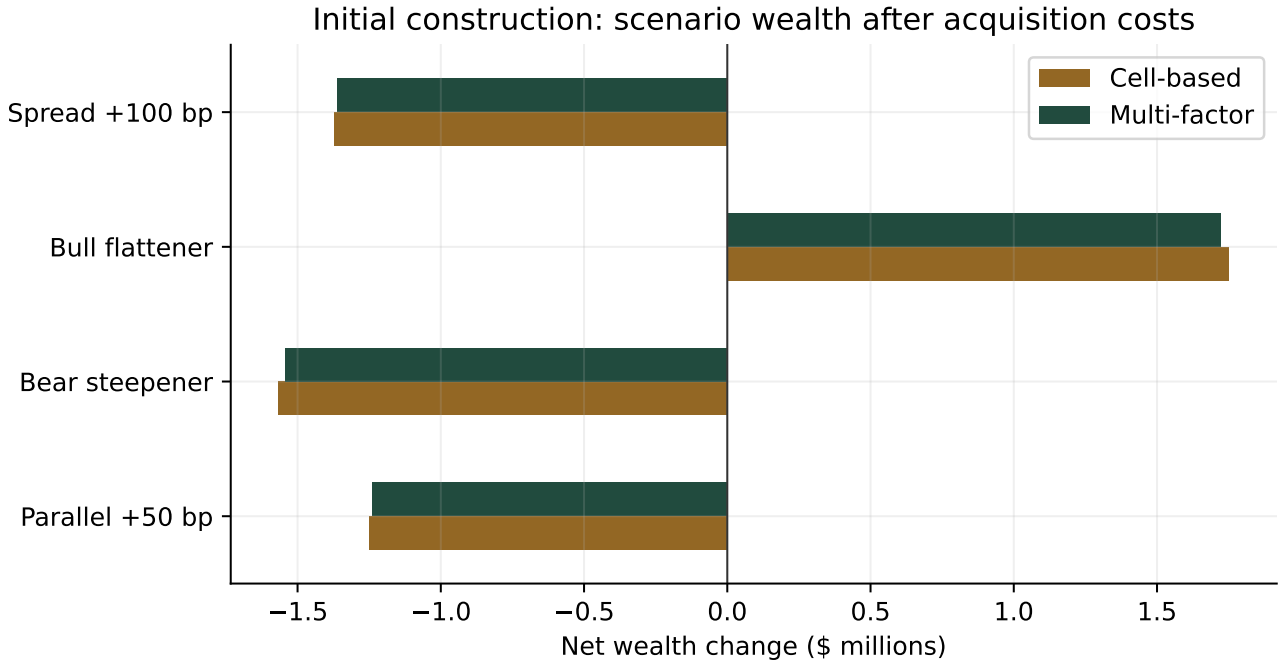
Gross price P&L applies to the actual amount invested in each bond sleeve. Net wealth change is terminal wealth minus initial cash: subtract the one-time purchase costs from that gross P&L. The benchmark is a costless reference index with the full $50 million invested. Neither table adds annual carry to an instantaneous shock.

Gross price P&L ($ millions)

Scenario	Benchmark	Cell-based	Multi-factor
Parallel +50 bp	-1.209	-1.221	-1.211
Bear steepener	-1.512	-1.538	-1.512
Bull flattener	1.752	1.781	1.750
Spread +100 bp	-1.320	-1.343	-1.331

Net wealth change after purchase costs ($ millions)

Scenario	Benchmark	Cell-based	Multi-factor
Parallel +50 bp	-1.209	-1.250	-1.241
Bear steepener	-1.512	-1.567	-1.542
Bull flattener	1.752	1.752	1.721
Spread +100 bp	-1.320	-1.372	-1.360



9 Explore construction choices and factor shocks

The HTML tools use the calculated allocations and risk matrix above. The PDF contains the corresponding tables and figures. These controls perform local calculations; they do not request market data or solve a new optimization.

9.1 Move between the two feasible construction plans

A weighted mixture of the two plans remains feasible because all mandate constraints are linear in bond-sleeve weights. The risk model evaluates each mixture; it need not be a linear average of the two endpoint tracking errors.

9.2 Change the four market factors explicitly

The default is a +50 bp parallel Treasury move and +25 bp common-spread widening. Set only one Treasury slider to a nonzero value to isolate a key-rate shock. Set all three Treasury sliders to the same value for a parallel shift.

10 Validate and interpret the decision

The calculations compare all four analytical factor sensitivities and parallel convexity with numerical repricing, including zero exposure beyond a short bond's maturity. They also check that weights sum to one, no short positions appear, every common mandate limit passes, exact cells match under the cell method, the covariance matrices are positive definite, and acquisition budgets reconcile. The multi-factor portfolio is also checked to have no more modeled TE than the feasible cell portfolio under the same common mandate.

The cell method provides a transparent benchmark-bucket match and a low-cost selection rule. The multi-factor method explicitly accounts for co-movement and residual risk. Neither method is guaranteed to outperform. Before investing, verify security data, issuer relationships, executable prices, settlement costs, and feasible lot sizes, then rerun the constraints on the rounded purchase plan.

11 Mini project: raise the yield-proxy requirement

Raise the minimum weighted-yield increment from **2 bp to 5 bp above the benchmark**, keeping every other input and limit fixed.

1. Determine whether exact cell matching can meet the revised requirement.
2. Find the maximum attainable yield proxy under exact cells and the other limits.
3. Construct a multi-factor target under the revised floor, if feasible.
4. Report its annual TE, duration, sector weights, and purchase-budget reconciliation.
5. Explain which requirement would need to change if a method is infeasible.

Do not label an infeasible solution a portfolio. Do not interpret a higher yield proxy as a guaranteed expected excess return.

Worked solution: feasibility and the revised construction

The greatest exact-cell yield increment is **2.713 bp**, below the revised 5 bp floor. Exact cell matching is infeasible under the revised mandate.

Target	Yield proxy (%)	Duration (years)	Annual TE (bp)
Multi-factor, +5 bp floor	5.137	4.942	8.819

Target	Treasury (%)	Corporate (%)	Largest bond (%)
Multi-factor, +5 bp floor	25.00	50.00	15.43

Bond	Weight (%)	Initial buy $(m)\|Cost()$	
UST 2Y	9.614	4.804	480.436
UST 5Y	5.385	2.691	538.181
UST 10Y	10.000	4.997	1499.170
Agency 3Y	9.571	4.782	1434.736
Agency 7Y	15.429	7.710	3855.026
Bank A 3Y	6.691	3.344	2006.206
Bank B 5Y	9.232	4.613	3229.167
Utility A 7Y	8.585	4.290	3860.824
Industrial A 10Y	5.023	2.510	3012.111
Industrial B 5Y	9.707	4.851	3880.548
Telecom A 10Y	4.042	2.020	2827.829
Consumer A 7Y	6.720	3.358	3357.776

Bond values of **$49,970,017.99** plus costs of **$29,982.01** use exactly **$50,000,000.00** of initial cash.

The exact-cell approach would require a lower yield floor, different cell budgets, a broader investable universe, or another relaxed constraint. The multi-factor approach meets the revised floor by using its freedom across cells, with greater active risk than its original 2 bp target. That is a construction tradeoff, not evidence of a free improvement.